PRACTICE · EDITION 1.0

Generative AI Question and Answer Bank

Questions and model answers based on the Generative AI Introduction and Study Notes

What this bank contains

- Part A: 35 short-answer questions, grouped by chapter
- Part B: 11 long-answer questions, including diagram-based questions
- Part C: 16 multiple-choice questions with an answer key

Companion to the Introduction study notes · Chapters 1 to 9
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HOW TO USE THIS BANK

Answer each question from memory first, then compare with the model answer. Check any gap against the matching chapter of the Introduction study notes.

Long-answer questions refer to Figures 1 to 6 in the study notes. Study those figures first.

Part C includes an answer key at the end. Attempt all questions before checking it.

Part A Short-Answer Questions

35 questions, grouped by chapter. Each answer is kept to what the study notes state.

Chapters 1 and 2: Foundations of AI

Q1. How are artificial intelligence, machine learning, deep learning and generative AI related?

They are nested. AI is the broadest field. Machine learning is a subset of AI that learns patterns from data. Deep learning is a subset of machine learning that uses neural networks with many layers. Generative AI is a class of models, usually deep learning models, whose output is new content.

Q2. What is the difference between supervised and self-supervised learning?

Supervised learning uses labelled examples, where each input comes with a known output. Self-supervised learning creates its own labels from the data, for example by hiding a word and training the model to predict it. Self-supervised learning is used in the pre-training of large language models.

Q3. What is reinforcement learning? Give one example.

It is learning from rewards received after actions taken in an environment. An example is a game-playing agent that improves its moves based on whether it wins.

Q4. What does a discriminative model learn?

It learns the boundary between categories, so it can predict a label or value from an input. It does not learn how to create new data.

Q5. Why can a generative model create new data while a discriminative model cannot?

A generative model learns how the data itself is distributed, so it can draw new samples from that distribution. A discriminative model only learns to separate classes or predict a value from an input.

Chapter 3: What Is Generative AI?

Q6. Define generative AI.

Generative AI refers to models that learn the underlying patterns of a training dataset and use them to produce new, original-looking outputs such as text, images, audio, video, code or 3D objects. The outputs resemble the training data without being copies of it.

Q7. List the four stages that most generative systems follow.

The four stages are: collect data; train a model; run inference by sampling from the learned probabilities for a prompt; and refine the model through fine-tuning and alignment.

Q8. What are parameters, and why do they matter?

Parameters, also called weights, are the numbers a model learns during training. They encode the patterns in the data. Large models have billions of parameters.

Q9. What is sampling, and why can the same prompt give different outputs?

Sampling means choosing an output at random according to a probability distribution. Because the model chooses from probabilities rather than looking up one stored answer, the same prompt can produce different outputs on different runs.

Chapter 4: Major Families of Generative Models

Q10. Who introduced GANs, and what are their two networks?

Ian Goodfellow and colleagues introduced GANs in 2014. The two networks are the generator, which turns random noise into a sample, and the discriminator, which tries to tell real samples from generated ones.

Q11. What is mode collapse?

Mode collapse is a weakness of GANs in which the generator produces only a few kinds of output instead of the full variety present in the training data.

Q12. In a VAE, what do μ and σ represent, and how is the latent code z obtained?

μ is the mean and σ is the standard deviation of the distribution that the encoder produces in the latent space. The latent code is sampled as $z = \mu + \sigma \cdot \varepsilon$, where ε is random noise drawn from a standard normal distribution.

Q13. Why are VAE outputs often described as blurrier than GAN or diffusion outputs?

The study notes list blurriness as a weakness of VAEs. Their outputs are often blurrier than those from GANs or diffusion models, even though VAE training is more stable.

Q14. Describe the forward and reverse processes in a diffusion model.

In the forward process, a small amount of Gaussian noise is added to an image over many steps until only random noise remains. This process is fixed and involves no learning. In the reverse process, a neural network learns to predict and remove the noise one step at a time. To generate an image, the model starts from pure noise and denoises it.

Q15. What is the main weakness of diffusion models?

Sampling needs many denoising steps, so generation is slow and compute-heavy.

Q16. What is an autoregressive model?

It is a model that generates a sequence one token at a time. Each new token is predicted from all the tokens that come before it. Large language models are autoregressive.

Q17. Name the four landmark papers in the notes, with their years and contributions.

2013: Kingma and Welling introduced the variational autoencoder. 2014: Goodfellow and colleagues introduced GANs. 2017: Vaswani and colleagues introduced the Transformer in Attention Is All You Need. 2020: Ho, Jain and Abbeel made diffusion models practical for high-quality image generation.

Chapter 5: The Transformer Architecture

Q18. What is a token, and what is an embedding?

A token is a unit of text, such as a word, part of a word or a character, produced by the tokenizer. An embedding is a vector of numbers that represents the meaning of a token. Tokens with related meanings have embeddings that are close together.

Q19. Write the self-attention formula and explain the roles of Q, K and V.

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})\, V$. Each token's query Q is compared with every token's key K to measure relevance. The resulting weights, after the softmax, are used to combine the value vectors V.

Q20. Why is the score divided by the square root of d_k?

d_k is the size of the key vectors. Dividing by its square root keeps the scores in a stable range, which keeps the softmax well behaved.

Q21. Why does a Transformer need positional encoding?

Attention on its own ignores the order of words. Positional encoding adds information about each token's position, so the model can distinguish the sequence of words.

Q22. What does multi-head attention do?

It runs several attention operations in parallel. Each head can track a different kind of relationship between tokens.

Q23. List the steps an LLM follows to generate one token.

The prompt is tokenized. The tokens are embedded and given positional information. They pass through the stacked Transformer blocks. A softmax produces a probability for each token in the vocabulary. One token is sampled, appended to the input, and the loop repeats.

Q24. What is the effect of temperature, and what do top-k and top-p do?

A low temperature makes output focused and predictable. A high temperature makes it more varied and creative but raises the risk of errors. Top-k and top-p sampling restrict the choice to a shortlist of the most likely tokens.

Chapter 6: Training, Adaptation and Prompting

Q25. What is the difference between pre-training and fine-tuning?

Pre-training teaches a model general patterns from a very large corpus using a self-supervised objective, and it is the most expensive stage. Fine-tuning continues training on a smaller, targeted dataset to specialise the model's behaviour.

Q26. What is RLHF?

Reinforcement learning from human feedback. People compare or rank model outputs. A reward model learns these preferences, and the language model is tuned to produce answers people judge helpful, honest and safe.

Q27. Which adaptation methods change the model's weights?

Pre-training and fine-tuning change the weights. Prompt engineering and retrieval-augmented generation do not, because they change only the input or the context supplied to the model.

Q28. What is retrieval-augmented generation (RAG), and how does it reduce hallucination?

In RAG, the system searches a document store for passages relevant to the question and places them in the prompt. The model then answers from the retrieved text. Grounding the answer in real sources reduces hallucination and lets the system use current or private information without retraining.

Q29. What is few-shot prompting?

Few-shot prompting gives the model a few worked examples in the prompt before the real question, so it can follow the pattern of the examples.

Chapters 7 and 8: Applications and Risks

Q30. Give two applications of generative AI in software development and two in education.

Software development: code generation and explanation, bug finding, writing unit tests and documentation. Education: personalised tutoring, quiz and question generation, and revision notes. Any two from each list is a correct answer.

Q31. What is hallucination, and name two safeguards against it.

Hallucination is fluent output that is factually wrong or unsupported by the input. Two safeguards are grounding the model with retrieved sources (RAG) and asking for citations. Human verification of important facts is also essential.

Q32. Name one risk linked to deepfakes and a suitable mitigation.

The risk is misinformation: realistic fake images, audio or video can mislead people. Mitigations include content labelling, watermarking and provenance standards.

Q33. Why should confidential or personal data not be pasted into public AI tools?

Data entered into a prompt may be stored or exposed by the system, so sensitive information could leak. Users should avoid sharing confidential data, use approved tools and follow access controls.

Q34. What three-part structure should you use when answering a question about a risk?

Name the risk, give a concrete example of it, and state one mitigation.

Chapter 9: Summary and Glossary

Q35. What is a latent space?

A latent space is a compressed, learned space in which data is represented by a small number of values. In a VAE it is smooth, so nearby points produce similar outputs.

Part B Long-Answer Questions

11 questions. Each is worth 5 marks. Model answers are shown in full so you can check the structure and the level of detail.

L1. Explain the hierarchy of AI, machine learning, deep learning and generative AI. Use Figure 1 in your answer.

5 marks

Artificial intelligence is the broad field of building systems that perform tasks requiring intelligence. Machine learning is a subset of AI in which systems learn patterns from data rather than following hand-written rules. Deep learning is a subset of machine learning that uses neural networks with many layers, such as CNNs, RNNs and Transformers. Generative AI is a class of models, usually deep learning models, whose output is new content. Figure 1 shows the four fields as nested circles, with generative AI at the centre. Examples of generative models include GPT-style language models, GANs, VAEs and diffusion models.

L2. Compare discriminative and generative models. Refer to Figure 2 and include a comparison table.

5 marks

A discriminative model learns the boundary between categories and predicts a label from an input. Figure 2 shows a boundary separating two classes. A generative model learns the distribution of the data and can draw new samples from it, shown in Figure 2 as new points (stars) placed within the learned shape. The main differences are: discriminative models answer 'which class?', while generative models answer 'what does the data look like, and can we create more of it?'. Discriminative models cannot create new data; generative models can. Examples of discriminative models include logistic regression and support vector machines. Examples of generative models include GANs, VAEs, diffusion models and GPT-style language models.

L3. Describe how a GAN is trained. Refer to Figure 3, and state one strength and one weakness.

5 marks

A GAN has two networks. The generator takes random noise and produces a fake sample. The discriminator receives real training data and generated samples and tries to classify each one as real or fake. Training is a game: the discriminator is updated to catch fakes more accurately, and the generator is updated using the discriminator's feedback so that its fakes look more real. Figure 3 shows this loop. A strength is that GANs produce sharp, realistic images and generate a sample in a single forward pass. A weakness is unstable training, which can lead to mode collapse, where the generator produces only a few kinds of output.

L4. Explain the variational autoencoder. Refer to Figure 4 and explain the role of the noise term ε.

5 marks

A VAE has an encoder, a latent space and a decoder. The encoder maps an input to a distribution in the latent space, described by a mean μ and a standard deviation σ. A latent code z is sampled from that distribution using $z = \mu + \sigma \cdot \varepsilon$, where ε is random noise from a standard normal distribution. The decoder reconstructs the input from z. The noise term makes the sampling step differentiable, so the network can be trained with gradient descent while still producing a random sample. VAEs have stable training and a smooth latent space, which is useful for interpolation. Their outputs are often blurrier than those from GANs or diffusion models.

L5. Explain how diffusion models generate an image. Refer to Figure 5.

5 marks

A diffusion model is trained using two processes. In the forward process, Gaussian noise is added to a clean image in many small steps until only random noise remains. This process is fixed and involves no learning. In the reverse process, a neural network learns to predict and remove the noise step by step. To generate a new image, the model starts from pure random noise and applies the learned denoising steps repeatedly. Figure 5 shows the clean image becoming progressively noisier on the forward path and being recovered on the reverse path. Diffusion models produce high-quality, diverse outputs with stable training, but sampling requires many steps, which makes generation slow and compute-heavy.

L6. Explain how a transformer-based language model generates a response. Refer to Figure 6 and the self-attention formula.

5 marks

The prompt is first converted into tokens by a tokenizer. Each token is mapped to an embedding, and positional encoding is added so the model knows the order of the words. The sequence passes through several stacked Transformer blocks. In each block, self-attention computes $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})\, V$, so every token can weigh the other tokens. A softmax at the output produces a probability for every token in the vocabulary. One token is sampled according to these probabilities, appended to the input, and the whole loop repeats until a stop condition is met. Figure 6 shows this loop. Because each new token depends on all earlier tokens, the output is generated one token at a time.

L7. Compare the four major families of generative models. Use a table in your answer.

5 marks

The four families are GANs, VAEs, diffusion models and autoregressive Transformers. GANs use a generator and a discriminator that compete; they are sharp and fast but unstable. VAEs encode inputs into a latent distribution and decode them; they are stable and have a smooth latent space but produce blurrier output. Diffusion models learn to reverse a gradual noising process; they give high-quality, diverse results but are slow to sample. Autoregressive Transformers predict the next token; they offer flexible, strong language skills but can hallucinate and are expensive to train and run. A table with columns for core idea, typical output, strength and weakness is the best format for this answer.

L8. Describe the stages that shape a language model's behaviour, and when each technique is used.

5 marks

Pre-training learns general patterns from a very large corpus using next-token prediction. It is the most expensive stage. Fine-tuning then continues training on a smaller, targeted dataset to specialise the model. Reinforcement learning from human feedback tunes the model so its answers match human preferences for helpfulness, honesty and safety. Prompt engineering improves results by changing the input, without changing weights. Retrieval-augmented generation supplies relevant documents at query time, which reduces hallucination and allows up-to-date or private information to be used without retraining. A sensible choice is to start with prompt engineering, add RAG when specific documents are needed, and fine-tune only when a specialised behaviour cannot be achieved otherwise.

L9. Identify at least four risks of generative AI, and give an example and a mitigation for each.

5 marks

Hallucination: confident but false output, such as a made-up reference. Mitigation: grounding with RAG, citations and human checking. Bias: models reflect unfair patterns in their training data, for example stereotyped descriptions. Mitigation: diverse data, bias testing and review. Misinformation and deepfakes: realistic fake images or audio can mislead people. Mitigation: labelling, watermarking and provenance standards. Privacy: sensitive data entered into a prompt can be exposed. Mitigation: avoid sharing confidential data and use approved tools. Other risks include copyright uncertainty, environmental cost and over-reliance that weakens learning.

L10. Explain temperature and sampling. Which setting would you choose for generating code, and which for brainstorming ideas?

5 marks

Temperature controls how random the choice of the next token is. A low temperature makes the model select the most likely tokens, so output is focused and predictable. A high temperature gives more variety, which suits brainstorming, but increases the chance of errors. Top-k and top-p sampling restrict the choice to a shortlist of likely tokens. For generating code, a low temperature is preferred, because a single correct and reliable answer is usually needed. For brainstorming ideas, a higher temperature is preferred, because variety is more valuable than precision.

L11. Explain how the discriminator and the generator in a GAN depend on each other, and why this makes training difficult.

5 marks

The two networks are adversaries. As the discriminator becomes better at detecting fakes, the generator receives a different training signal and must adapt. As the generator improves, the discriminator must adapt in turn. Because each network's success depends on the other's current state, the process is a minimax game rather than a simple optimisation. If one network becomes too strong, the other can stop learning, and the generator can collapse to a narrow set of outputs. This instability is why GAN training is considered difficult.

Part C Multiple-Choice Questions

16 questions. Each is worth 1 mark. Choose one option.

M1. Which field is the broadest of the four?

A) Machine learning
B) Artificial intelligence
C) Deep learning
D) Generative AI

M2. Which type of learning uses labelled examples?

A) Unsupervised learning
B) Supervised learning
C) Reinforcement learning
D) Self-supervised learning

M3. In a GAN, which network tries to distinguish real samples from generated ones?

A) Generator
B) Encoder
C) Discriminator
D) Decoder

M4. In a VAE, how is the latent code z obtained?

A) $\mu \times \sigma \times \varepsilon$
B) $\mu + \sigma \cdot \varepsilon$
C) $\mu - \sigma$
D) $\varepsilon \div \sigma$

M5. What kind of process is the forward process of a diffusion model?

A) Learned by a neural network
B) Fixed, adding noise step by step
C) Driven by human rewards
D) Random selection of pixels

M6. Which architecture is built on self-attention?

A) Convolutional neural network
B) Transformer
C) Support vector machine
D) Decision tree

M7. Why is positional encoding used in a Transformer?

A) To compress the data
B) To provide word-order information
C) To remove noise
D) To set the temperature

M8. What does a low temperature produce?

A) More random output
B) Focused, predictable output
C) Longer output
D) Faster training

M9. Which technique does not change the model's weights?

A) Fine-tuning
B) Pre-training
C) RLHF
D) Prompt engineering

M10. Which technique uses human rankings of model outputs?

A) RLHF
B) RAG
C) Pre-training
D) Diffusion

M11. Which technique retrieves documents to ground an answer?

A) RAG
B) VAE
C) GAN
D) Softmax

M12. A well-known weakness of GANs is:

A) Slow sampling
B) Mode collapse
C) Blurry outputs
D) Tokenization errors

M13. The Transformer architecture was introduced in which year?

A) 2013
B) 2014
C) 2017
D) 2020

M14. Fluent output that is factually wrong is called:

A) A token
B) An embedding
C) Hallucination
D) A latent space

M15. The softmax function is used to:

A) Clean the training data
B) Turn scores into probabilities

C) Compress images
D) Label the dataset

M16. Which is a safeguard against deepfake misinformation?

A) Watermarking and provenance standards
B) Raising the temperature to 2.0
C) Adding more parameters
D) Ignoring content labels

Answer Key for Part C

Question	Answer	Correct option
M1	**B**	Artificial intelligence
M2	**B**	Supervised learning
M3	**C**	Discriminator
M4	**B**	$\mu + \sigma \cdot \varepsilon$
M5	**B**	Fixed, adding noise step by step
M6	**B**	Transformer
M7	**B**	To provide word-order information
M8	**B**	Focused, predictable output
M9	**D**	Prompt engineering
M10	**A**	RLHF
M11	**A**	RAG
M12	**B**	Mode collapse
M13	**C**	2017
M14	**C**	Hallucination
M15	**B**	Turn scores into probabilities
M16	**A**	Watermarking and provenance standards

STUDY TIP

Missed a question? Return to the chapter it belongs to, reread the matching figure, and write the answer again in your own words without looking at the model answer.